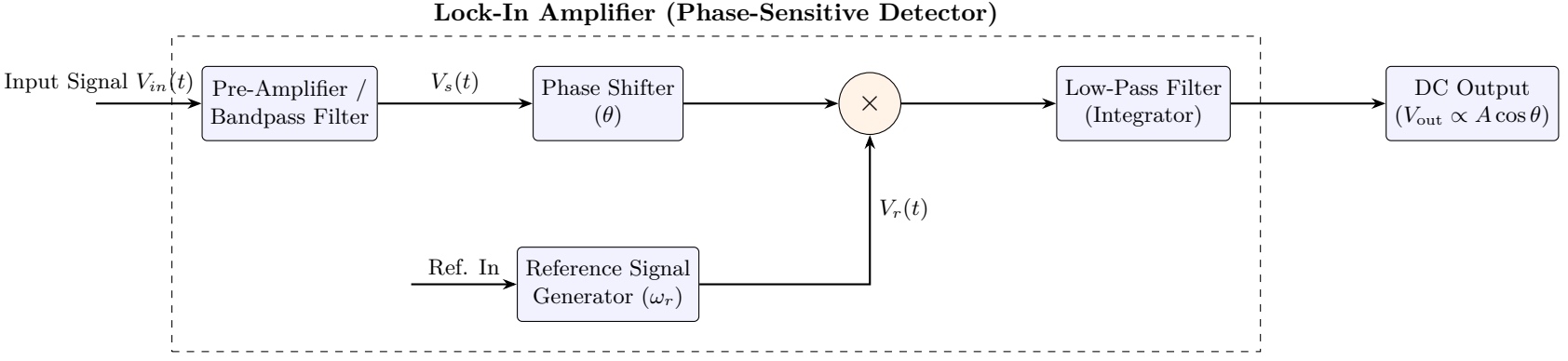
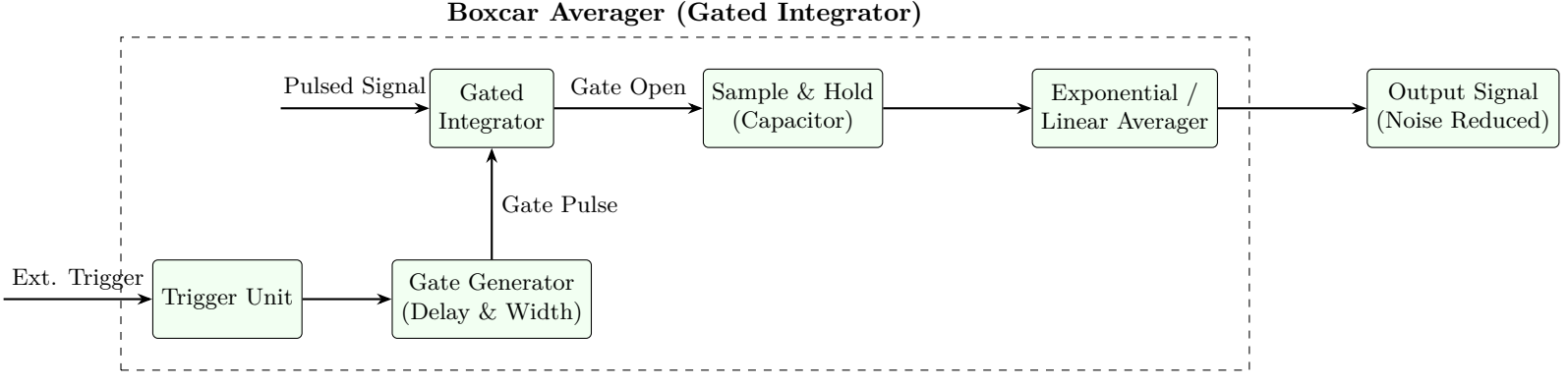

Lock-In Amplifier (Phase-Sensitive Detector)
Input Signal $V_{in}(t)$
Pre-Amplifier / Bandpass Filter
$V_s(t)$
Phase Shifter (θ)
$\times$
Low-Pass Filter (Integrator)
DC Output ($V_{\text{out}} \propto A\cos\theta$)
$V_r(t)$
Ref. In
Reference Signal Generator (ω_r)


Boxcar Averager (Gated Integrator)
Pulsed Signal
Gated Integrator
Gate Open
Sample & Hold (Capacitor)
Exponential / Linear Averager
Output Signal (Noise Reduced)
Gate Pulse
Ext. Trigger
Trigger Unit
Gate Generator (Delay & Width)